Advanced Computer Networks — Consolidated Final Study Note

Everything for the comprehensive final in one place: the midterm recap (delays, HTTP/DNS, subnetting), the Transport Layer (Ch 3), and the Link Layer (Ch 6), each brought to life with **animated, interactive simulations**. Built from Kurose & Ross, *Top-Down Approach* (9th ed.) and the course slides.

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PART 0 · CUMULATIVE

Midterm Recap — Delays, Application Layer, Network Layer

0.1 · Physical delays: transmission vs propagation

Total nodal delay is the sum of four pieces. Only **queueing** varies with load; the rest are fixed for a given link and packet.

```
d_nodal = d_proc + d_queue + d_trans + d_prop

d_proc  : processing (check errors, pick output link)   ~ constant, µs
d_queue : waiting in the output buffer                  VARIABLE (load)
d_trans : L / R   push all bits onto the link           fixed for L, R
d_prop  : d / s   bits travel down the link             fixed for the link
```

The classic confusion: **transmission delay** is the time to *push* every bit of a packet onto the wire (depends on packet length L and link rate R). **Propagation delay** is the time for one bit to *travel* the distance (depends on distance d and signal speed s only). Watch it:

Transmission vs Propagation

d_trans = L/R	**0.800 ms**
d_prop = d/s	**15.000 ms**
total (last bit arrives)	**15.800 ms**

bits being pushed (transmission) bit travelling (propagation)

Interactive simulation (open the HTML file to run): bits are pushed onto the wire over d_trans = L/R, then travel the link over d_prop = d/s.

EXAM TRAP

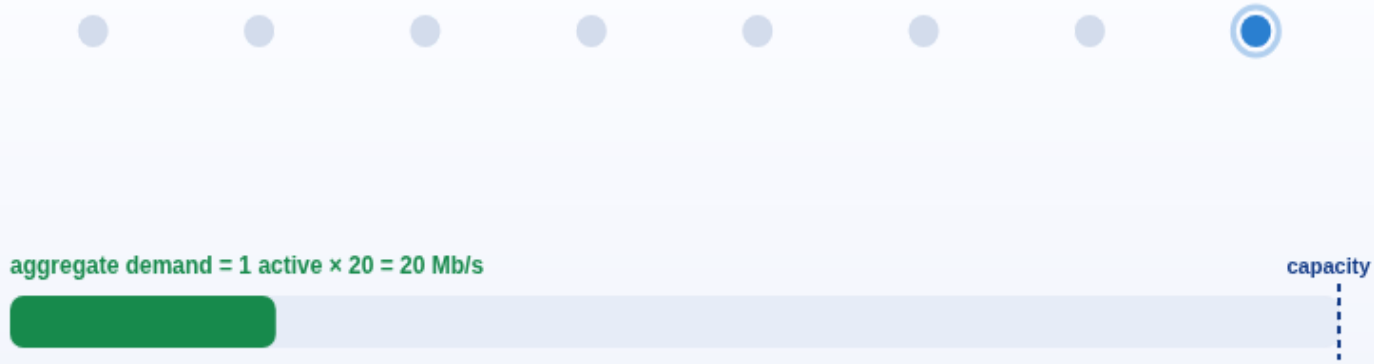
d_prop does **not** depend on L or R. A bigger packet or a faster link changes d_trans, never d_prop. Store-and-forward through one switch: $d = L/R_1 + L/R_2$ (the switch must receive the whole packet before resending).

```
Store-and-forward (1 switch): d = L/R₁ + L/R₂
Queueing (n packets ahead, x bits of current one sent): d_queue = ((n+1)·L − x) / R
```

0.2 · Packet switching & statistical multiplexing

Packet switching **shares** a link on demand (statistical multiplexing); circuit switching **reserves** a slice per user even when idle. Sharing works because users are rarely all active at once.

Statistical multiplexing — will it congest?



active now	**1 / 8**
E[active] = N·p	**1.2**
status	**OK (shared safely)**

Interactive: N users each blink active p% of the time at r Mb/s on a shared C Mb/s link.
The aggregate bar turns red when demand exceeds capacity.

```
Circuit switching:  N_max = C / r         (each user reserves r forever)
Packet switching:   active users X ~ Binomial(N, p),  E[X] = N·p
                    congestion when X·r > C  →  usually rare
```

WORKED

1 Gb/s link, r = 100 Mb/s per active user, active 10% of time. Circuit: N_max = 1000/100 = **10**. Packet with 35 users: E[X] = 3.5, and P(more than 10 active) < 0.0004 — so 35 users share safely.

0.3 · Throughput & the bottleneck link

End-to-end throughput is set by the **slowest link** on the path. A chain of pipes flows only as fast as its narrowest segment.

Bottleneck throughput = min(R_1, R_2, R_3)

throughput = min(R_1,R_2,R_3)	**3 Mb/s**
time for a 10 MB file	**26.7 s**

Interactive: three links of adjustable rate; the flowing bit-rate equals the minimum (the bottleneck).

```
throughput = min(R₁, R₂, …, Rₙ)      transfer time of file F = F / throughput
```

0.4 · Application layer — HTTP, DNS, web caching

A **socket = IP address : port**. The IP finds the host (building); the 16-bit port finds the process (apartment). Applications pick TCP or UDP:

	TCP	UDP
Connection	3-way handshake	connectionless
Reliability	reliable, in-order, retransmits	best-effort, may lose/reorder
Flow / congestion control	yes / yes	no / no
Header	20+ bytes	8 bytes
Typical apps	HTTP, HTTPS, SMTP, FTP, SSH	DNS, VoIP, live video, games

HTTP: persistent vs non-persistent

HTTP runs over TCP and is **stateless** (cookies add state on top). Non-persistent opens a fresh TCP connection per object (~2·RTT each); persistent reuses one connection.

HTTP page load — persistent vs non-persistent (RTT ladder)

round trips	**4 RTT**
total load time	**160 ms**
vs the other mode	**non-persistent ≈ 6 RTT**

TCP handshake HTTP request response

Interactive time-sequence ladder: toggle persistent/non-persistent and object count to see how round trips add up. Non-persistent repeats the TCP handshake per object.

Web caching — access-link math

```
Offered traffic  T = λ · L            Utilisation U = T / C   (as U→1, delay→∞)
With cache hit rate h, only misses use the link:  T' = (1−h)·λ·L
Avg delay = (1−h)·d_miss + h·d_hit
```

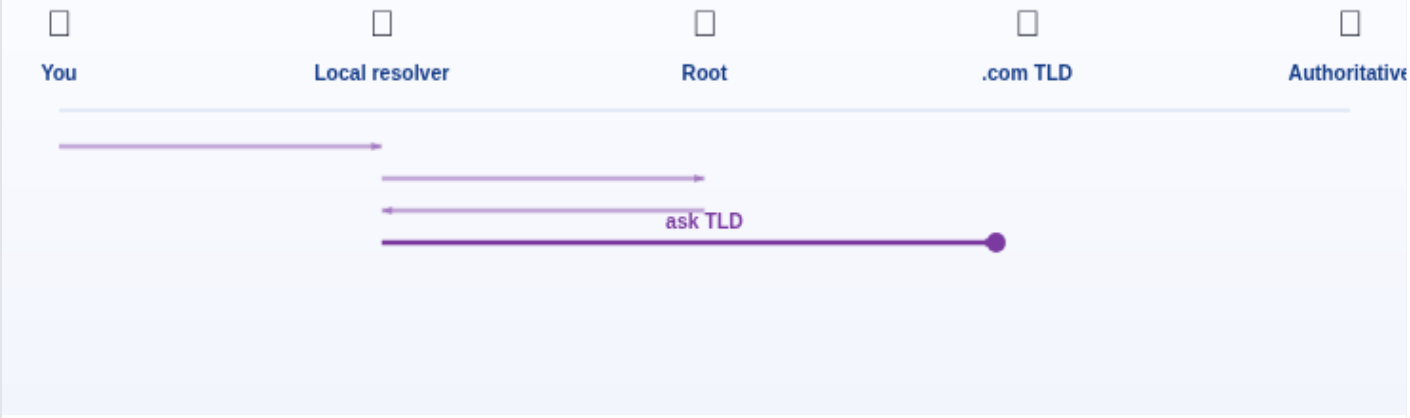
WORKED

Access link 1.54 Mb/s, object 100 kb, 15 req/s, RTT 2 s. T = 1.5 Mb/s → U = 0.97 (minutes of delay). Cache h = 0.4 → T' = 0.9 Mb/s → U' = 0.58, avg delay ≈ 1.2 s.

DNS — the Internet's phone book

Translates names → IPs. It is **distributed & hierarchical** (root → TLD → authoritative) to avoid a single point of failure, huge load, distant lookups, and unmaintainability. Records: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), NS (name server).

DNS resolution walk (iterative)



mode	**cache miss — hierarchical walk**
step	**ask TLD**

Interactive: watch a query walk local resolver → root → TLD → authoritative and return the IP; a cache hit short-circuits it.

0.5 · Network layer — IP addressing, subnetting, VLSM, NAT

IPv4 is **32 bits** = four octets (0–255), split into a network part (left) and host part (right) by the mask / CIDR `/m`. Try the calculator — it does the block-size trick and shows the binary split:

Subnet calculator (block-size method)

IP address: 172.16.35.123

Prefix /m: 20

Network bits shown in blue, host bits in amber. Uses the block-size method.

Interactive: enter an IP and prefix; get network, broadcast, host range, usable hosts, and a binary network/host split.

```
host bits n = 32 − m       usable hosts = 2ⁿ − 2 (minus network + broadcast)
block size = 256 − (last non-255 mask octet)     network = block start
broadcast = next block start − 1
```

Classful (identify by first octet): A 0–127 (/8), B 128–191 (/16), C 192–223 (/24), D 224–239 (multicast), E 240–255 (reserved). Private: 10/8, 172.16/12, 192.168/16.

VLSM — variable length subnet masks

Give each subnet its own mask, just big enough. **Recipe:** sort requirements largest → smallest; pick the smallest n with $2^n - 2 \geq \text{hosts}$; assign the block; start the next subnet right after the previous broadcast.

LAN	Need	2^n-2	Prefix	Network	Broadcast
LAN1	100	126 (n=7)	/25	192.168.10.0	192.168.10.127
LAN2	50	62 (n=6)	/26	192.168.10.128	192.168.10.191
LAN3	20	30 (n=5)	/27	192.168.10.192	192.168.10.223

EXAM TRAP

Allocate the **largest subnet first**, or a small one will land on a boundary the big one needs. Always subtract 2 (network + broadcast).

NAT — network address translation

Lets a whole private network share **one** public IPv4 address. The router rewrites (source IP, source port) outbound and reverses it inbound, tracking each mapping in a NAT table. Pros: conserves IPv4, hides topology. Cons: breaks end-to-end addressing.

PART A · CHAPTER 3

The Transport Layer

A.1 · Multiplexing & demultiplexing

The transport layer extends host-to-host (IP) delivery to **process-to-process**. The receiver *demultiplexes* each segment to the right socket using header fields.

- **UDP demux = 2-tuple** (dest IP, dest port). Same dest port from different sources → same socket.
- **TCP demux = 4-tuple** (src IP, src port, dst IP, dst port). Each client gets its own connection socket even on port 80.

Interactive: segments arrive and are routed to sockets. Toggle UDP/TCP to see 2-tuple vs 4-tuple demultiplexing.

EXAM TRAP

For UDP the source is just a return address and does not change which socket receives data. For TCP all four fields matter.

A.2 · Reliable data transfer & pipelining

We build reliability on an unreliable channel step by step. **Sequence numbers** catch duplicates; **timers** catch loss. rdt3.0 is *stop-and-wait* — one packet in flight — which wastes the link.

```
Stop-and-wait utilisation:  U = (L/R) / (RTT + L/R)
Example: 1 Gb/s, RTT 30 ms, L 8000 b → L/R = 8 µs → U ≈ 0.027%  (terrible!)
```

Interactive: compare one-in-flight stop-and-wait to a pipelined sender (window N).
Utilisation bar fills much more with pipelining.

Go-Back-N vs Selective Repeat

	Go-Back-N	Selective Repeat
ACKs	cumulative	individual
Timers	one (oldest unACKed)	one per unACKed packet
On loss	resend packet n and all higher	resend only the lost packet
Receiver buffer	none (discard out-of-order)	buffers out-of-order
Window rule	$N \le 2^k - 1$	$N \le 2^{k-1}$ (half the seq space)

Interactive: toggle GBN/SR, then drop a packet. GBN re-sends everything from the loss; SR re-sends only the missing one.

A.3 · TCP — connection, sequence numbers, flow control

TCP is a reliable, in-order **byte stream**: point-to-point, full-duplex, pipelined, with cumulative ACKs. Seq # = byte-stream number of the first data byte; ACK # = next byte expected.

Interactive: play the SYN → SYN-ACK → ACK exchange with sequence numbers; both sides reach ESTABLISHED.

```
EstimatedRTT = (1−α)·EstimatedRTT + α·SampleRTT        (α = 0.125)
DevRTT       = (1−β)·DevRTT + β·|SampleRTT − EstimatedRTT|  (β = 0.25)
TimeoutInterval = EstimatedRTT + 4·DevRTT
```

Fast retransmit: 3 duplicate ACKs → resend the missing segment before the timer fires.

Flow control (protects the receiver)

The receiver advertises free buffer space in **rwnd**; the sender caps in-flight data to rwnd so the receive buffer never overflows.

Interactive: sender rate vs app read rate fill/drain the receive buffer; rwnd (advertised free space) throttles the sender.

EXAM TRAP

Flow control (rwnd) protects the **receiver**. Congestion control (cwnd) protects the **network**. Different mechanisms.

A.4 · Congestion control — slow start & AIMD

TCP probes for bandwidth with a congestion window **cwnd**; it sends `min(cwnd, rwnd)`. Slow start ramps exponentially to **ssthresh**, then congestion avoidance adds linearly (AIMD). On loss, `ssthresh = cwnd/2`.

Interactive/animated cwnd-vs-round graph: exponential slow start, linear congestion avoidance, halving on triple-dup-ACK, reset to 1 on timeout.

READ A P40 GRAPH

Doubling per round = **slow start**. +1 per round = **congestion avoidance**. Window **halved** on loss = triple-duplicate-ACK (Reno). Window **dropped to 1** = timeout. On any loss, ssthresh = cwnd/2.

```
AIMD:  additive increase +1 MSS/RTT ;  multiplicative decrease ×½ on 3-dup-ACK
avg TCP throughput ≈ (3/4)·W / RTT          fairness: K flows → ~R/K each
```

PART B · CHAPTER 6

The Link Layer & LANs

B.1 · Error detection & correction

The sender adds redundant EDC bits. **2-D parity** can not only detect but *correct* a single-bit error by pinpointing its row and column. Flip a bit and watch it get located:

Interactive: click any data bit to flip it; the mismatched row and column parity pinpoint the error for correction.

CRC is stronger: pick r check bits R so that ⟨D,R⟩ is divisible by generator G (mod-2). The receiver divides by G; a nonzero remainder means error. Detects all bursts ≤ r bits (used by Ethernet, WiFi).

Interactive: enter data D and generator G; step through the XOR long-division to get the CRC remainder R.

B.2 · Multiple access — CSMA/CD

On a shared broadcast medium, two simultaneous transmissions **collide**. CSMA = listen before transmit; CSMA/CD also aborts on collision detection and backs off. Collisions still happen because of propagation delay (a node may not yet hear the other).

Interactive: two nodes sense and transmit; overlapping signals collide, a jam is sent, and each backs off a random K·512 bit-times.

```
Slotted ALOHA max efficiency = 1/e ≈ 37%     Pure ALOHA = 1/(2e) ≈ 18%
CSMA/CD backoff: after m-th collision pick K ∈ {0…2ᵐ−1}, wait K·512 bit-times
Collision if two start within d_prop:  yes when d_prop < L/R
```

B.3 · MAC addressing & ARP

A 48-bit **MAC address** (flat, in NIC ROM) moves a frame between interfaces on the same subnet. **ARP** maps a known IP → its MAC: broadcast the query, get a unicast reply.

Interactive: A broadcasts "who has IP?" to all nodes; only the owner replies unicast, and A caches the mapping.

CORE IDEA

Across the Internet, **IP addresses stay fixed end-to-end** while **MAC addresses are rewritten at every hop**. ARP query = broadcast; ARP reply = unicast.

B.4 · Ethernet switches — self-learning

A switch is a transparent, plug-and-play, store-and-forward layer-2 device. It **learns** which MAC lives on which port from the source address of arriving frames, then forwards (or floods if the destination is unknown).

Interactive: send frames between hosts; the switch table learns source ports, floods unknown destinations, and forwards known ones on a single port.

Switch	Router
Layer 2, examines MAC	Layer 3, examines IP
Table by flooding + learning	Table by routing algorithms
Does not change TTL	Decrements TTL each hop

VLANs split one physical switch into isolated broadcast domains; 802.1Q adds a 4-byte tag with a 12-bit VLAN ID → up to $\mathbf{2^{12} = 4096}$ VLANs.

B.5 · A day in the life of a web request

The grand synthesis, laptop → www.google.com:

1. **DHCP** (broadcast, UDP/IP/Ethernet) → laptop gets its IP, gateway IP, DNS server IP.
2. **ARP** → laptop learns the first-hop router's MAC.
3. **DNS** query (UDP) → returns google's IP.
4. **TCP** 3-way handshake to the web server.
5. **HTTP** GET over TCP → server returns the page → browser renders.

COMPANION

For the full hop-by-hop packet journey (every router rewriting MACs, TTL counting down, IP constant), open `browsing_the_internet.html` in this folder.

PART C · ASSIGNED

Solved Exam Questions

Chapter 3 — Review questions

R3. How is a UDP socket fully identified? A TCP socket? Difference?

UDP: a 2-tuple (dest IP, dest port). **TCP:** a 4-tuple (src IP, src port, dest IP, dest port). UDP ignores the source for demux; TCP uses all four, so different clients on port 80 get different sockets.

R4. Why might a developer choose UDP over TCP?

No congestion throttling (rate-sensitive media), no handshake RTT, no per-client connection state (more clients), smaller header — and the app can add exactly the reliability it needs (e.g. HTTP/3 over UDP).

R14. True or False?

#	Statement	Answer	Why
a	B won't ACK A because it can't piggyback	**False**	TCP sends pure ACK segments anytime.
b	rwnd never changes	**False**	It changes as the buffer fills/drains.
c	Unacked bytes ≤ receive-buffer size	**True**	Flow control caps in-flight ≤ rwnd.
d	Next seq # is necessarily m+1	**False**	Next = m + bytes of data.
e	Header has an rwnd field	**True**	The 16-bit receive-window field.
f	SampleRTT 1 s ⇒ Timeout ≥ 1 s	**False**	Timeout uses smoothed EstRTT + 4·DevRTT.
g	seq 38 + 4 bytes ⇒ ACK is 42	**False**	42 is what B sends back, not A's own ACK.

R15. Two segments: seq 90 then seq 110.

(a) Data in first = 110 − 90 = **20 bytes**. (b) First lost, second arrives → cumulative ACK still requests byte **90** (a duplicate ACK).

R18. On timeout, ssthresh = ½ its previous value — T/F?

False. ssthresh = ½ of the current *cwnd*, not of the previous ssthresh. On a timeout, cwnd then resets to 1 MSS and slow start restarts.

Chapter 3 — Problems

P1. Two Telnet clients A, B → Server S (port 23).

Segment	Source port	Dest port
a. A → S	467 (any ephemeral)	23
b. B → S	513 (any ephemeral)	23
c. S → A	23	467
d. S → B	23	513

(e) Different hosts → source ports **may match** (IPs differ, so the 4-tuples differ). (f) Same host → source ports **must differ**.

P27. B has received through byte 126; A sends 80 then 40 bytes, first seq = 127.

- (a) First segment: seq **127**, src 302, dst 80.
- (b) Second: seq **207** (=127+80), 40 bytes.
- (c) First arrives first → ACK **207** (src 80, dst 302).
- (d) Second arrives first (gap) → ACK still **127**.

```
First lost, second arrives, then retransmit:
A --seq=127,80B--X (lost)
A --seq=207,40B--> arrives out of order
           <--ACK=127 (gap ACK)
(timeout / 3 dup-ACKs)
A --seq=127,80B--> gap filled
           <--ACK=247   (127+80+40 — cumulative)
```

P28. GBN cumulative-ACK robustness.

TCP acknowledges the next expected in-order byte, so one ACK can cover several earlier segments. If ACK1/ACK2 are lost but ACK3 (higher, cumulative) arrives, the sender advances past all three — no retransmission. Retransmit only on timeout or 3 dup-ACKs.

P40. cwnd-vs-round graph (standard: rounds 1–26, ssthresh 32 then 21).

- (a) Slow start: rounds **[1,6]** and **[23,26]** (exponential).
- (b) Congestion avoidance: **[6,16]** and **[17,22]** (linear).
- (c) After round 16: halved, stays >1 → **triple-dup-ACK**.
- (d) After round 22: drops to 1 → **timeout**.
- (e) Initial ssthresh = **32** (slow start ends at cwnd 32).
- (f) After round 18: ssthresh = 42/2 = **21**.
- (h) Loss after round 26 by 3-dup-ACK → cwnd and ssthresh both = **W/2**.

Chapter 6 — Review questions

R1. Transportation analogy: what is the frame?

The **transportation mode** (car, bus, train, plane) carrying the passenger (datagram) over one leg — as a datagram rides inside a frame across one link.

R2. If every link were reliable, is TCP reliability redundant?

No. Per-link reliability only covers one link. Datagrams can still arrive *out of order* (different routes) or be *lost* (routing loops, buffer overflow, equipment failure). TCP still provides the ordered, loss-free end-to-end byte stream.

R3. Link-layer services; which also in IP? TCP?

Service	In IP?	In TCP?
Framing	✓ (encapsulation)	✓
Link access	—	—
Reliable delivery	✗	✓
Flow control	✗	✓
Error detection	✓ (header checksum)	✓
Error correction	✗	✗
Full duplex	—	✓

R4. Two nodes transmit length-L at rate R; collision if d_prop < L/R?

Yes. If d_prop < L/R the sender is still transmitting when the other's signal arrives, so the signals overlap on the medium → collision.

R10. A → B unicast on a broadcast LAN; does C process/pass up? Broadcast?

Unicast to B's MAC: C's adapter processes then **discards** (dest MAC ≠ its own) — not passed up. Broadcast MAC: C **processes and passes up** to its network layer.

R11. Why ARP query broadcast but reply unicast?

The querier doesn't know which MAC owns the IP, so it must ask everyone (broadcast). The responder already learned the querier's MAC from the query, so it replies directly (unicast) without bothering the whole LAN.

REFERENCE

Formula Sheet & Exam Traps

```
Transmission delay   d_trans = L / R
Propagation delay    d_prop  = d / s
Store-and-forward    d = L/R₁ + L/R₂
Queueing (n ahead)   d_queue = ((n+1)·L − x) / R
Throughput           min(R₁,…,Rₙ);  file time = F / throughput
Stop-and-wait util   U = (L/R)/(RTT + L/R)
RTT estimate         EstRTT = (1−α)EstRTT + α·SampleRTT      (α=0.125)
RTT deviation        DevRTT = (1−β)DevRTT + β|SampleRTT−EstRTT| (β=0.25)
Timeout              EstRTT + 4·DevRTT
TCP throughput       ≈ (3/4)·W / RTT
SR window            N ≤ (seq space)/2
Slotted / Pure ALOHA 1/e ≈ 0.37  /  1/(2e) ≈ 0.18
CSMA/CD backoff      K ∈ {0…2ᵐ−1}; wait K·512 bit-times
Subnet hosts         2ⁿ − 2 ;  block = 256 − last mask octet
VLANs (802.1Q)       2¹² = 4096
MAC / IPv4 / IPv6    2⁴⁸ / 2³² / 2¹²⁸ addresses
```

Top exam traps

1. Flow control (rwnd) protects the receiver; congestion control (cwnd) protects the network.
2. Next TCP seq # = current seq # + bytes of data, **not** +1.
3. On loss ssthresh = cwnd/2. Timeout → cwnd = 1 MSS. 3-dup-ACK → cwnd = cwnd/2 (Reno).
4. UDP demux = 2-tuple; TCP demux = 4-tuple.
5. IP addresses stay fixed end-to-end; MAC addresses change every hop.
6. ARP query = broadcast; ARP reply = unicast.
7. Switch: unknown dest → flood; known dest same port → drop; else forward on one port.
8. Slow start = exponential; congestion avoidance = linear.
9. d_prop depends on distance only, never on L or R. Only d_queue is variable.
10. Throughput = min of link rates (bottleneck).
11. Usable hosts = $2^n - 2$; block size = 256 − last mask octet.
12. VLSM: allocate largest subnet first. Non-persistent HTTP ≈ 2·RTT per object.

Advanced Computer Networks — consolidated final study note. Content from Kurose & Ross, *Computer Networking: A Top-Down Approach* (9th ed.) and course slides. Simulations are illustrative. Built to be read on phone or printed to PDF.